MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – TEST (SET – C)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 11: Areas Related to Circles** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

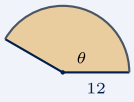
Official Evaluation Key | Maths Manthan Meerut

SECTION A: MULTIPLE CHOICE QUESTIONS MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The area of the sector of a circle of radius 12 cm is 60π cm^2. The central angle of this sector is: [1]

(A) 120° (B) 60° (C) 75° (D) 150°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]



Step 1: Formula and Given Values [0.5 Mark]
Radius $r = 12$ cm, Area of sector $= 60\pi$ cm^2.

$$\text{Area} = \frac{\theta}{360^\circ} \times \pi r^2 \implies \frac{\theta}{360^\circ} \times \pi(12)^2 = 60\pi$$

Step 2: Solving for θ [0.5 Mark]

$$\frac{\theta \times 144}{360} = 60 \implies \frac{\theta}{2.5} = 60 \implies \theta = 60 \times 2.5 = 150^\circ$$

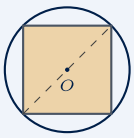
Correct Option: (D) 150°

Central angle $\theta = \mathbf{150^\circ}$

Q2. The area of a square inscribed in a circle of radius $2\sqrt{5}$ cm is: [1]

(A) 50 cm^2 (B) 100 cm^2 (C) 40 cm^2 (D) 80 cm^2

▶ STEP-BY-STEP SOLUTION: [1 Mark]



Step 1: Diagonal of Inscribed Square [0.5 Mark]

For a square inscribed in a circle of radius $r = 2\sqrt{5}$ cm:

$$\text{Diagonal } d = 2r = 2(2\sqrt{5}) = 4\sqrt{5} \text{ cm}$$

Step 2: Area of Square [0.5 Mark]

$$\text{Area} = \frac{d^2}{2} = \frac{(4\sqrt{5})^2}{2} = \frac{16 \times 5}{2} = \frac{80}{2} = 40 \text{ cm}^2$$

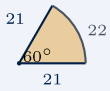
Correct Option: (C) 40 cm^2

$$\boxed{\text{Area of inscribed square} = \mathbf{40} \text{ cm}^{\mathbf{2}}}$$

Q3. The perimeter of a sector of a circle of radius 21 cm which subtends an angle of 60° at the centre of the circle is: [1]

(A) 22 cm (B) 43 cm (C) 64 cm (D) 462 cm

▶ STEP-BY-STEP SOLUTION: [1 Mark]



Step 1: Arc Length Calculation [0.5 Mark]

Radius $r = 21$ cm, $\theta = 60°$.

$$l = \frac{60°}{360°} \times 2\pi r = \frac{1}{6} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{6} \times 132 = 22 \text{ cm}$$

Step 2: Total Sector Perimeter [0.5 Mark]

$$\text{Perimeter} = l + 2r = 22 + 2(21) = 22 + 42 = 64 \text{ cm}$$

Correct Option: (C) 64 cm

$$\boxed{\text{Perimeter of sector} = \mathbf{64} \text{ cm}}$$

Q4. The numerical value of the area of a circle is equal to that of the perimeter of a semicircular disc, both having equal radius r. The radius is: [1]

(A) 1 unit (B) 2 units (C) $\left(1 + \frac{2}{\pi}\right)$ units (D) $\left(2 + \frac{2}{\pi}\right)$ units

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Step 1: Formulating the Equation [0.5 Mark]

$$\text{Area of circle} = \pi r^2$$

$$\text{Perimeter of semicircular disc} = \pi r + 2r = r(\pi + 2)$$

Step 2: Solving for r [0.5 Mark]

$$\pi r^2 = r(\pi + 2) \implies \pi r = \pi + 2 \implies r = \frac{\pi + 2}{\pi} = 1 + \frac{2}{\pi} \text{ units}$$

Correct Option: (C) $\left(1 + \frac{2}{\pi}\right)$ units

$$\boxed{\text{Radius } r = \left(\mathbf{1} + \frac{\mathbf{2}}{\pi}\right) \text{ units}}$$

Q5. If the perimeter of a square is equal to that of a circle, then the ratio of their areas is: [1]

(A) 7 : 22 (B) 11 : 14 (C) 22 : 7 (D) 14 : 11

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Equating Perimeters [0.5 Mark]

Let side of square be a and circle radius be r.

$$4a = 2\pi r \implies a = \frac{\pi r}{2}$$

Step 2: Ratio of Areas [0.5 Mark]

$$\frac{\text{Area(square)}}{\text{Area(circle)}} = \frac{a^2}{\pi r^2} = \frac{\pi^2 r^2/4}{\pi r^2} = \frac{\pi}{4} = \frac{22/7}{4} = \frac{22}{28} = \frac{11}{14} = 11 : 14$$

Correct Option: (B) 11 : 14

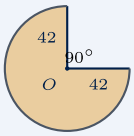
$$\boxed{\frac{\text{Area of square}}{\text{Area of circle}} = \mathbf{11 : 14}}$$

SECTION B: Short Answer Questions

MATHS MANTHAN | **(3 × 2 = 6 Marks)**

Q6. The shape of the top of a table is that of a major sector of a circle with centre O and central angle $\angle AOB = 90°$. If $OA = OB = 42$ cm, find the perimeter of the top of the table. $\left[\text{Use } \pi = \frac{22}{7}\right]$ [2]

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]



Step 1: Major Arc Length [1 Mark]

Radius $r = 42$ cm. Central angle of minor sector $= 90°$.

$$\text{Central angle of major sector } \theta' = 360° - 90° = 270°$$

$$\text{Major arc length } l = \frac{270°}{360°} \times 2\pi r = \frac{3}{4} \times 2 \times \frac{22}{7} \times 42 = \frac{3}{4} \times 264 = 198 \text{ cm}$$

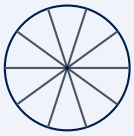
Step 2: Total Perimeter of Table Top [1 Mark]

$$\text{Perimeter} = l + OA + OB = 198 + 42 + 42 = 198 + 84 = 282 \text{ cm}$$

$$\boxed{\text{Perimeter of the top of the table} = \mathbf{282} \text{ cm}}$$

Q7. A brooch is made with silver wire in the form of a circle with diameter 35 mm. The wire is also used in making 5 diameters which divide the circle into 10 equal sectors. Find the total length of the silver wire used in making the brooch. $\left[\text{Use } \pi = \frac{22}{7}\right]$ **[2]**

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]



Step 1: Circumference of Circular Rim [1 Mark]

Diameter $d = 35$ mm.

$$\text{Circumference } C = \pi d = \frac{22}{7} \times 35 = 22 \times 5 = 110 \text{ mm}$$

Step 2: Total Wire Length [1 Mark]

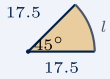
Wire is used for the circular boundary plus 5 diameters:

$$\text{Total length} = C + 5d = 110 + 5(35) = 110 + 175 = 285 \text{ mm}$$

Total length of silver wire = **285** mm

Q8. A circular pizza of diameter 35 cm is cut into 8 equal sectors. Find the perimeter of each slice of pizza. $\left[\text{Use } \pi = \frac{22}{7}\right]$ **[2]**

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]



Step 1: Arc Length of One Slice [1 Mark]

Diameter $d = 35$ cm $\implies$ radius $r = \frac{35}{2} = 17.5$ cm.

$$\text{Total Circumference } C = \pi d = \frac{22}{7} \times 35 = 110 \text{ cm}$$

$$\text{Arc length of 1 slice } l = \frac{C}{8} = \frac{110}{8} = 13.75 \text{ cm}$$

Step 2: Perimeter of Slice [1 Mark]

$$\text{Perimeter} = l + 2r = 13.75 + 2(17.5) = 13.75 + 35 = 48.75 \text{ cm}$$

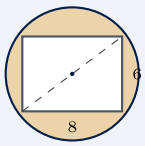
Perimeter of each slice = **48.75** cm

SECTION C: Long Answer Questions

MATHS MANTHAN | **(3 × 3 = 9 Marks)**

Q9. Reeti prepares a Rakhi for her brother Ronit. The Rakhi consists of a rectangle of length 8 cm and breadth 6 cm inscribed in a circle. Find the area of the shaded region (area of the circle excluding the rectangle). [Use $\pi = 3.14$] **[3]**

► STEP-BY-STEP SOLUTION: [3 Marks]



Step 1: Diagonal and Circle Radius [1 Mark]

The diagonal of an inscribed rectangle serves as the diameter of the circle.

$$\text{Diagonal } d = \sqrt{l^2 + b^2} = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10 \text{ cm}$$

$$\text{Radius of circle } r = \frac{d}{2} = \frac{10}{2} = 5 \text{ cm}$$

Step 2: Areas of Circle and Rectangle [1 Mark]

$$\text{Area of circle} = \pi r^2 = 3.14 \times 5^2 = 3.14 \times 25 = 78.5 \text{ cm}^2$$

$$\text{Area of rectangle} = l \times b = 8 \times 6 = 48 \text{ cm}^2$$

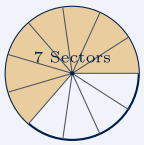
Step 3: Area of Shaded Region [1 Mark]

$$\text{Area of shaded region} = \text{Area of circle} - \text{Area of rectangle} = 78.5 - 48 = 30.5 \text{ cm}^2$$

$$\boxed{\text{Area of the shaded region} = \mathbf{30.5} \text{ cm}^2}$$

Q10. A circle of radius 12 cm is divided into 11 identical sectors. Find the total area of 7 such sectors. $\left[\text{Use } \pi = \frac{22}{7}\right]$ **[3]**

► STEP-BY-STEP SOLUTION: [3 Marks]



Step 1: Total Area of the Circle [1 Mark]

Radius $r = 12$ cm.

$$\text{Total Area} = \pi r^2 = \frac{22}{7} \times 12 \times 12 = \frac{22 \times 144}{7} = \frac{3168}{7} \text{ cm}^2$$

Step 2: Area of 1 Sector [1 Mark]

Since the circle is divided into 11 equal sectors:

$$\text{Area of 1 sector} = \frac{1}{11} \times \text{Total Area} = \frac{1}{11} \times \frac{3168}{7} = \frac{288}{7} \text{ cm}^2$$

Step 3: Area of 7 Sectors [1 Mark]

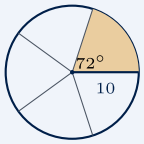
$$\text{Total Area of 7 sectors} = 7 \times \text{Area of 1 sector} = 7 \times \frac{288}{7} = 288 \text{ cm}^2$$

(Alternatively: Area $= \frac{7}{11} \times \left(\frac{22}{7} \times 144\right) = 2 \times 144 = 288$ *cm².)*

$$\boxed{\text{Total area of 7 such sectors} = \mathbf{288} \text{ cm}^2}$$

Q11. A circular disc of diameter 20 cm is divided equally into 5 sectors. Find the area and perimeter of one of the sectors. [Use $\pi = 3.14$] **[3]**

▶ **STEP-BY-STEP SOLUTION:** [3 Marks]



Step 1: Parameters of One Sector [0.5 Mark]

Diameter $d = 20$ cm $\implies$ radius $r = 10$ cm.

$$\text{Central angle } \theta = \frac{360^\circ}{5} = 72^\circ$$

Step 2: Area of One Sector [1 Mark]

$$\text{Area} = \frac{1}{5} \times \pi r^2 = \frac{1}{5} \times 3.14 \times 10^2 = \frac{314}{5} = 62.8 \text{ cm}^2$$

Step 3: Perimeter of One Sector [1.5 Marks]

$$\text{Arc length } l = \frac{1}{5} \times 2\pi r = \frac{1}{5} \times 2 \times 3.14 \times 10 = \frac{62.8}{5} = 12.56 \text{ cm}$$

$$\text{Perimeter} = l + 2r = 12.56 + 2(10) = 12.56 + 20 = 32.56 \text{ cm}$$

$$\boxed{\text{Area} = \mathbf{62.8} \text{ cm}^{\mathbf{2}}, \quad \text{Perimeter} = \mathbf{32.56} \text{ cm}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★